

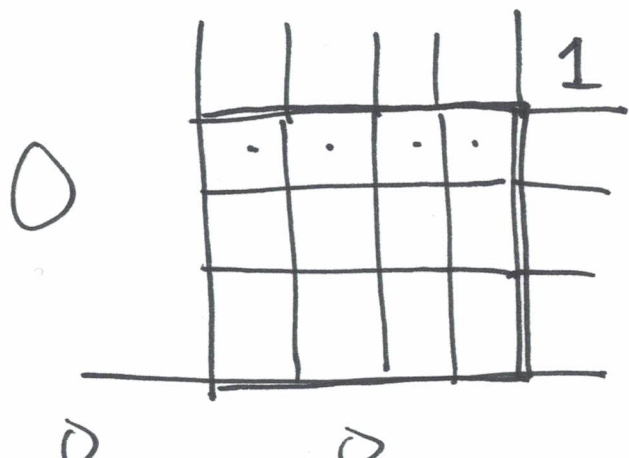
CONTINUOUS JOINT RANDOM VARIABLES

X AND Y ARE RANDOM VARIABLES (CONTINUOUS)

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) \quad \text{JOINT CDF}$$

$$f_{XY}(x, y) = \frac{d}{dx} \frac{d}{dy} F_{XY}(x, y) \quad \text{JOINT PDF}$$

$$\frac{\partial}{\partial x} \frac{\partial}{\partial y} F_{XY}(x, y)$$

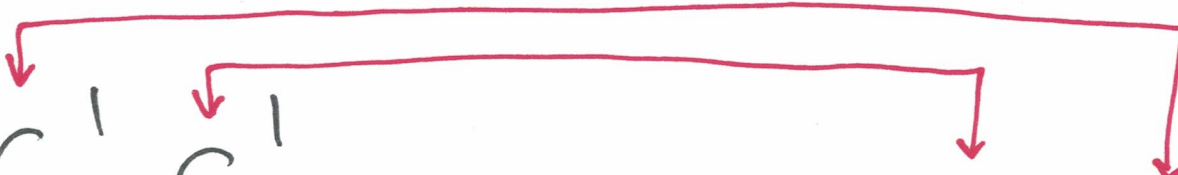


DISCRETE JOINT CDF
(LAST TIME)

CONSIDER $g(x, y) = c(2x + y^2)$

c A CONSTANT, DEFINED ON $x \in [0, 1]$
 $y \in [0, 1]$.

WHAT VALUE OF c MAKES THIS A VALID PDF?


$$\int_0^1 \int_0^1 c(2x + y^2) dx dy$$

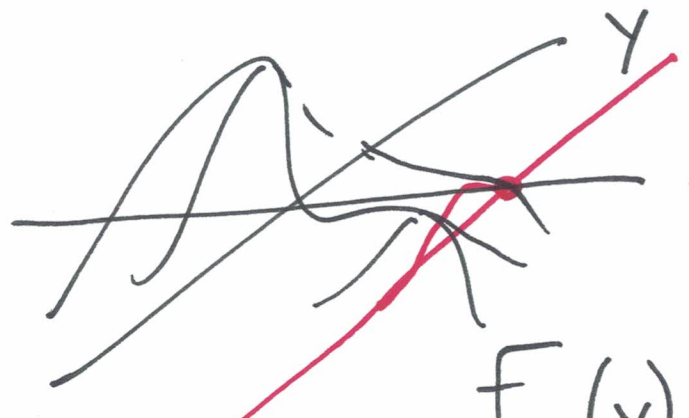
$$= c \int_0^1 \left[x^2 + xy^2 \right]_{x=0}^{x=1} dy = c \int_0^1 (1 + y^2) dy$$

$$= C \left(y + \frac{1}{3} y^3 \right) \Big|_{y=0}^{y=1} = \frac{4}{3} C \Rightarrow C = \frac{3}{4}.$$

MARGINAL PDF:

GIVEN JOINT PDF $f_{XY}(x, y)$

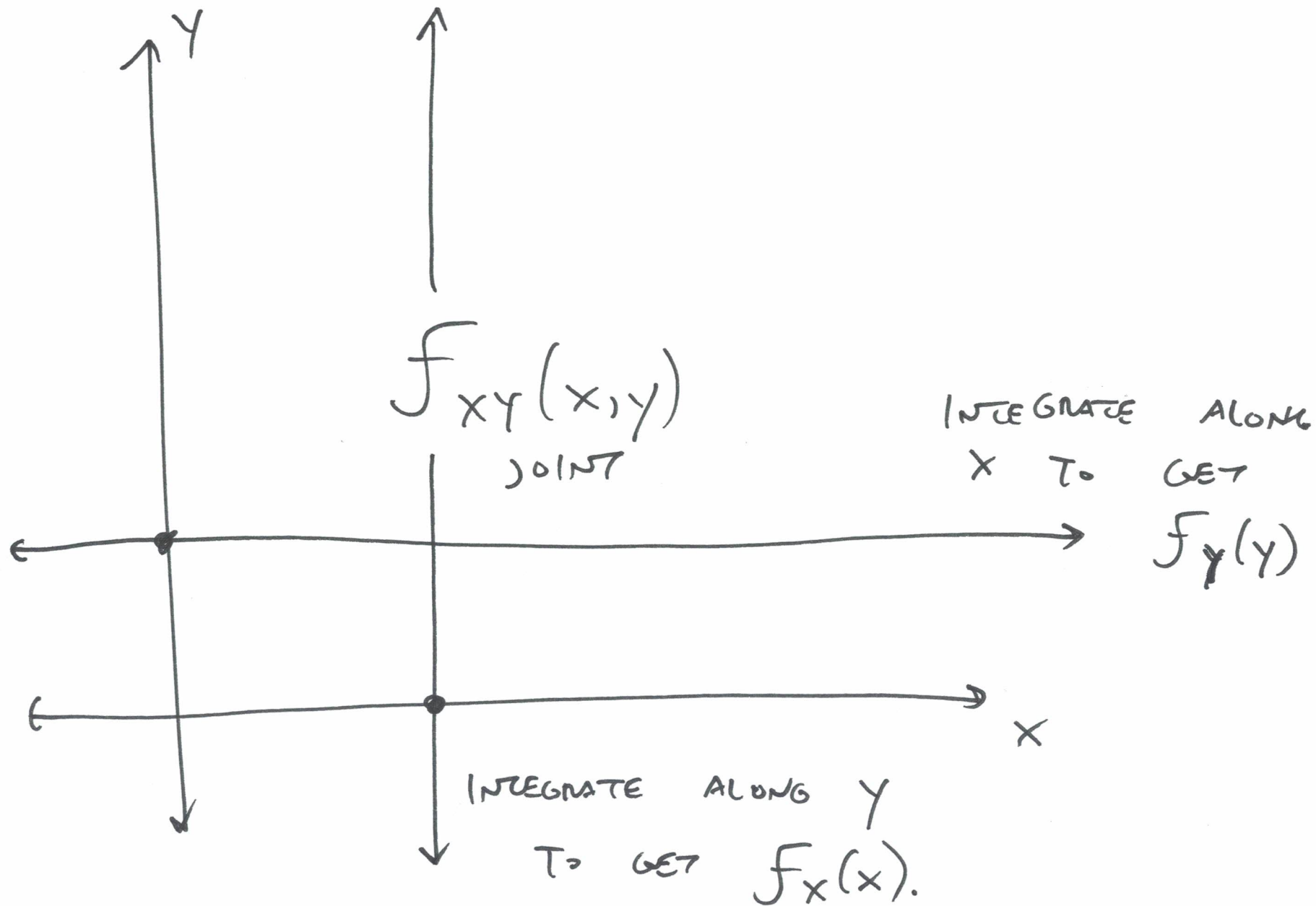
WE CAN DERIVE THE PDF OF
JUST ONE OR THE OTHER VARIABLE.



$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

← DON'T CARE ABOUT Y →



IN EXAMPLE, WHAT IS MARGINAL IN x ?

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, y) dy$$

$$= \int_0^1 \frac{3}{4} (2x + y^2) dy$$

$$= \left[\frac{3}{2} xy + \frac{1}{4} y^3 \right]_{y=0}^{y=1}$$

$$= \begin{cases} \frac{3}{2}x + \frac{1}{4} & x \in [0, 1] \\ 0 & \text{ELSE.} \end{cases}$$

SANITY CHECK: $\int_0^1 \left(\frac{3}{2}x + \frac{1}{4} \right) dx = \left[\frac{3}{4}x^2 + \frac{1}{4}x \right]_{x=0}^{x=1} = 1. \checkmark$

MARGINAL IN y :

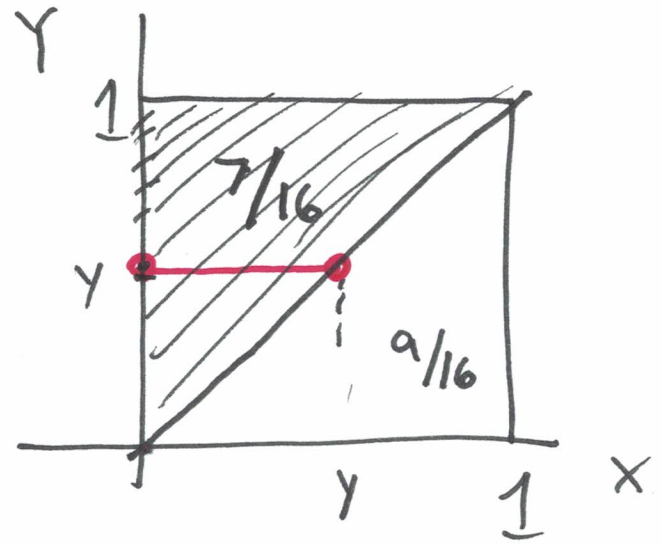
$$F_Y(y) = \int_0^1 \frac{3}{4}(2x + y^2) dx$$

$$= \left[\frac{3}{4}x^2 + \frac{3}{4}y^2x \right]_{x=0}^{x=1}$$

$$= \begin{cases} \frac{3}{4} + \frac{3}{4}y^2 & y \in [0, 1] \\ 0 & \text{ELSE} \end{cases}$$

COMPUTE $P(X \leq Y)$.

$$= \int_0^1 \int_0^Y \left(\frac{3}{2}x + \frac{3}{4}y^2 \right) dx dy$$



$$= \int_0^1 \left[\frac{3}{4}x^2 + \frac{3}{4}xy^2 \right]_{x=0}^{x=y} dy$$

$$= \int_0^1 \left(\frac{3}{4}y^2 + \frac{3}{4}y^3 \right) dy = \left[\frac{1}{4}y^3 + \frac{3}{16}y^4 \right]_{y=0}^{y=1}$$

$$= \boxed{\frac{7}{16}}.$$

WE TALKED ABOUT THE INDEPENDENCE
OF EVENTS: ^{EVENTS} A AND B ARE INDEPENDENT IF

$$P(A \cap B) = P(A) P(B)$$

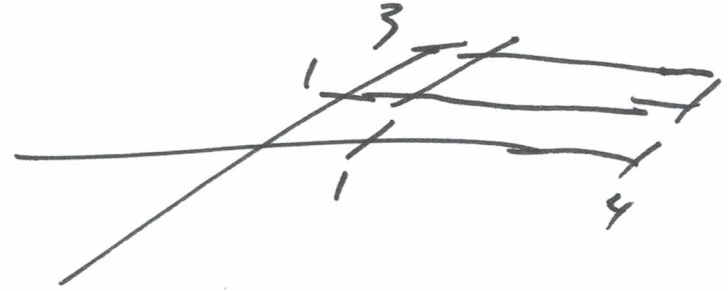
TWO RANDOM VARIABLES X AND Y
ARE INDEPENDENT IF:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$\text{JOINT} = (\text{MARGINAL IN } x)(\text{MARGINAL IN } y).$$

$$\text{IN EX: } \frac{3}{2}x + \frac{3}{4}y^2 \stackrel{?}{=} \left(\frac{3}{2}x + \frac{1}{4}\right) \left(\frac{3}{4} + \frac{3}{4}y^3\right) \quad \text{X}$$

$$f_{xy}(x,y) = \frac{1}{6} \quad \begin{array}{l} x \in [1,4] \\ y \in [1,3] \end{array}$$



$$= \left(\frac{1}{3} \quad x \in [1,4] \right) \left(\frac{1}{2} \quad y \in [1,3] \right)$$

$$f_{xy}(x,y) = \text{scribble} \quad x^2 y^3$$

$$\begin{array}{l} x \in \text{---} \\ y \in \text{---} \end{array}$$

$$= \left(\text{---} x^2 \right) \left(\text{---} y^3 \right)$$

SOME MORE ABOUT THE GAUSSIAN:

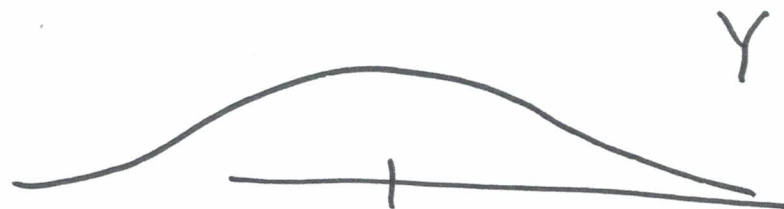
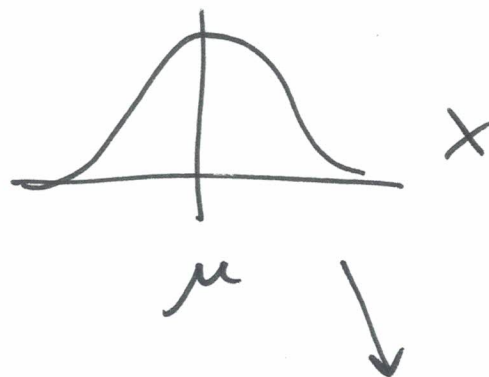
X IS GAUSSIAN WITH MEAN μ
VARIANCE σ^2 .

$$Y = aX + b.$$

Y IS STILL A GAUSSIAN:

$$\mu_Y = a\mu_X + b$$

$$\sigma_Y^2 = a^2 \sigma_X^2$$



JOINT GAUSSIAN RANDOM VARIABLE (2)

IS SPECIFIED BY:

μ_x, μ_y : MEANS IN X AND Y

σ_x^2, σ_y^2 : VARIANCES IN X AND Y

"RHO" ρ : CORRELATION COEFFICIENT.

$$f_{xy}(x,y) = \frac{1}{2\pi |\Sigma|^{1/2}} e^{-\frac{1}{2} ([x] - \mu)^T \Sigma^{-1} ([x] - \mu)}$$

$\uparrow$ DETERMINANT

$\mu = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$ INVER

COVARIANCE MATRIX $\Sigma = \begin{bmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{bmatrix}$

WRITING IT ALL OUT:

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_x)^2}{\sigma_x^2} - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \frac{(y-\mu_y)^2}{\sigma_y^2} \right]}$$

WE CAN SEE THAT IF $\rho = 0$,
THEN

$$f_{XY}(x,y) = f_X(x) f_Y(y)$$

GAUSSIAN
IN x

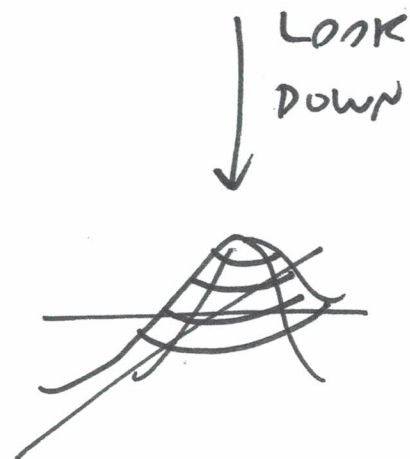
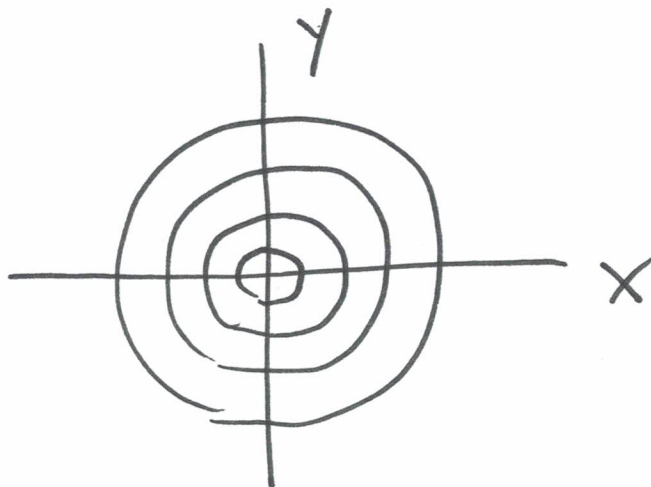
GAUSSIAN
IN y

THINK ABOUT GAUSSIAN PDF LIKE A
TOPOGRAPHICAL (HIKING) MAP:

$$\rho = 0$$

$$\sigma_x = \sigma_y = 1$$

$$\mu_x = \mu_y = \phi$$

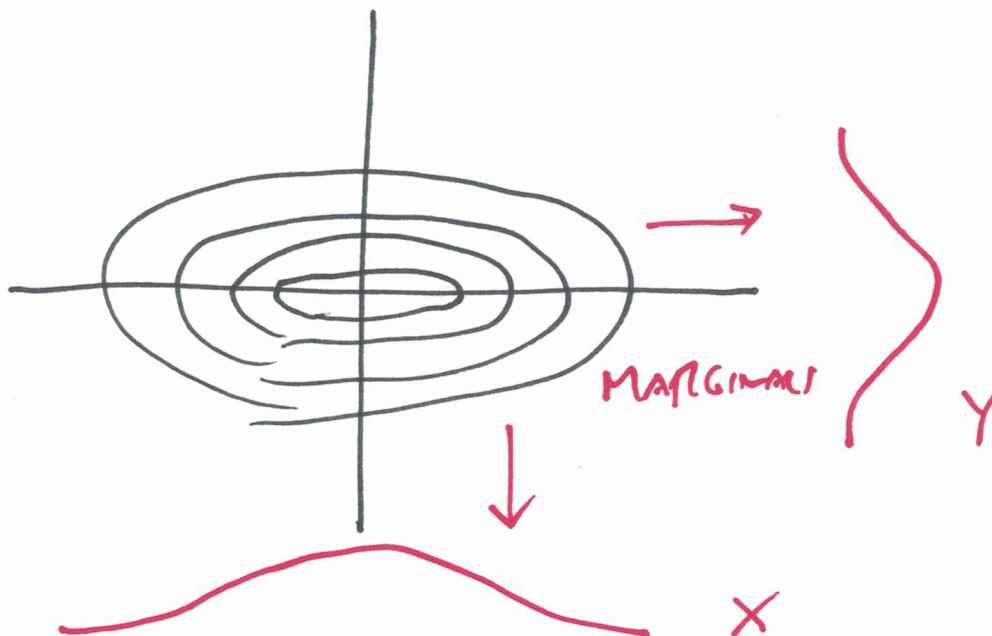


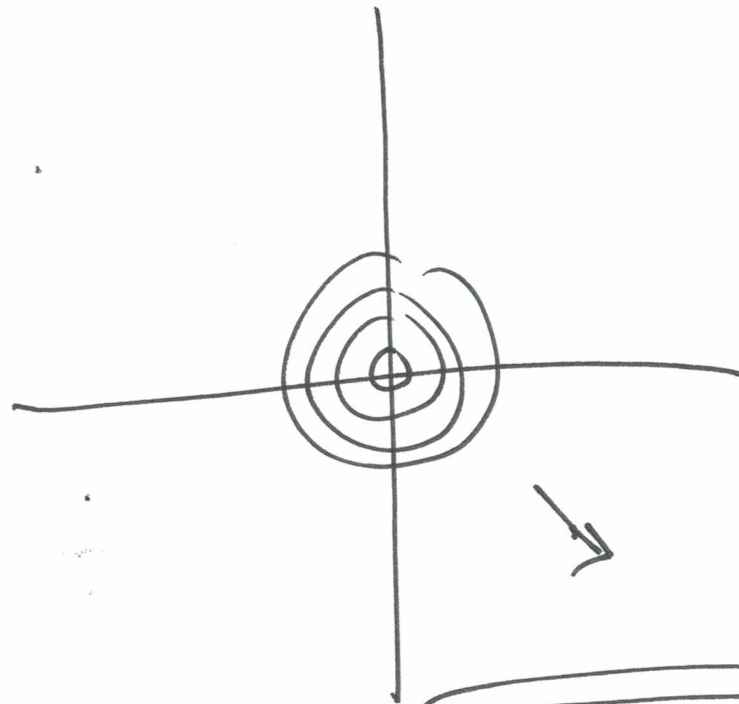
$$\rho = 0$$

$$\sigma_x = 2$$

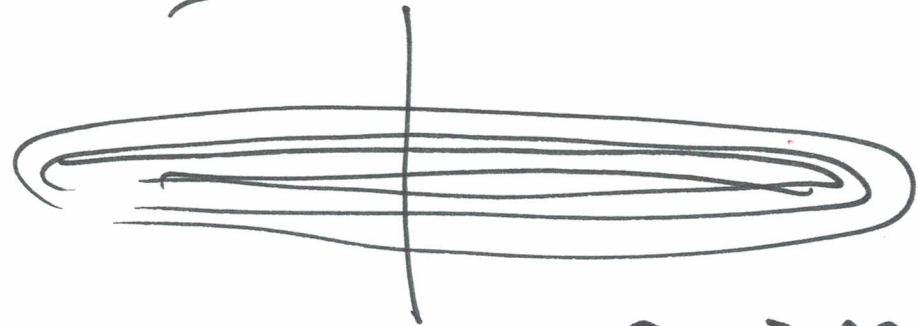
$$\sigma_y = 1$$

$$\mu_x = \mu_y = \phi$$





$$\sigma_x = \sigma_y = 1$$



$$\sigma_x \rightarrow \infty$$

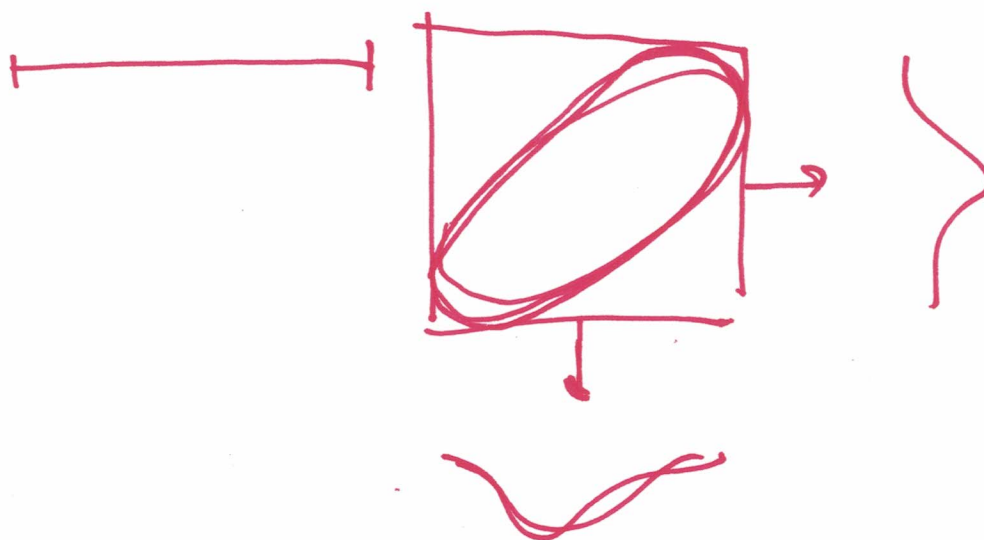
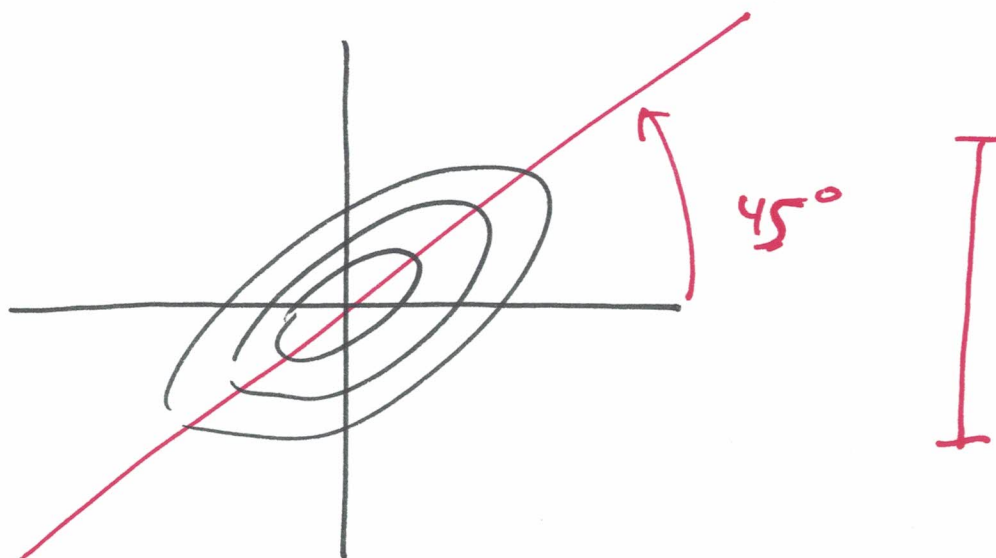
$$\sigma_y = 1$$

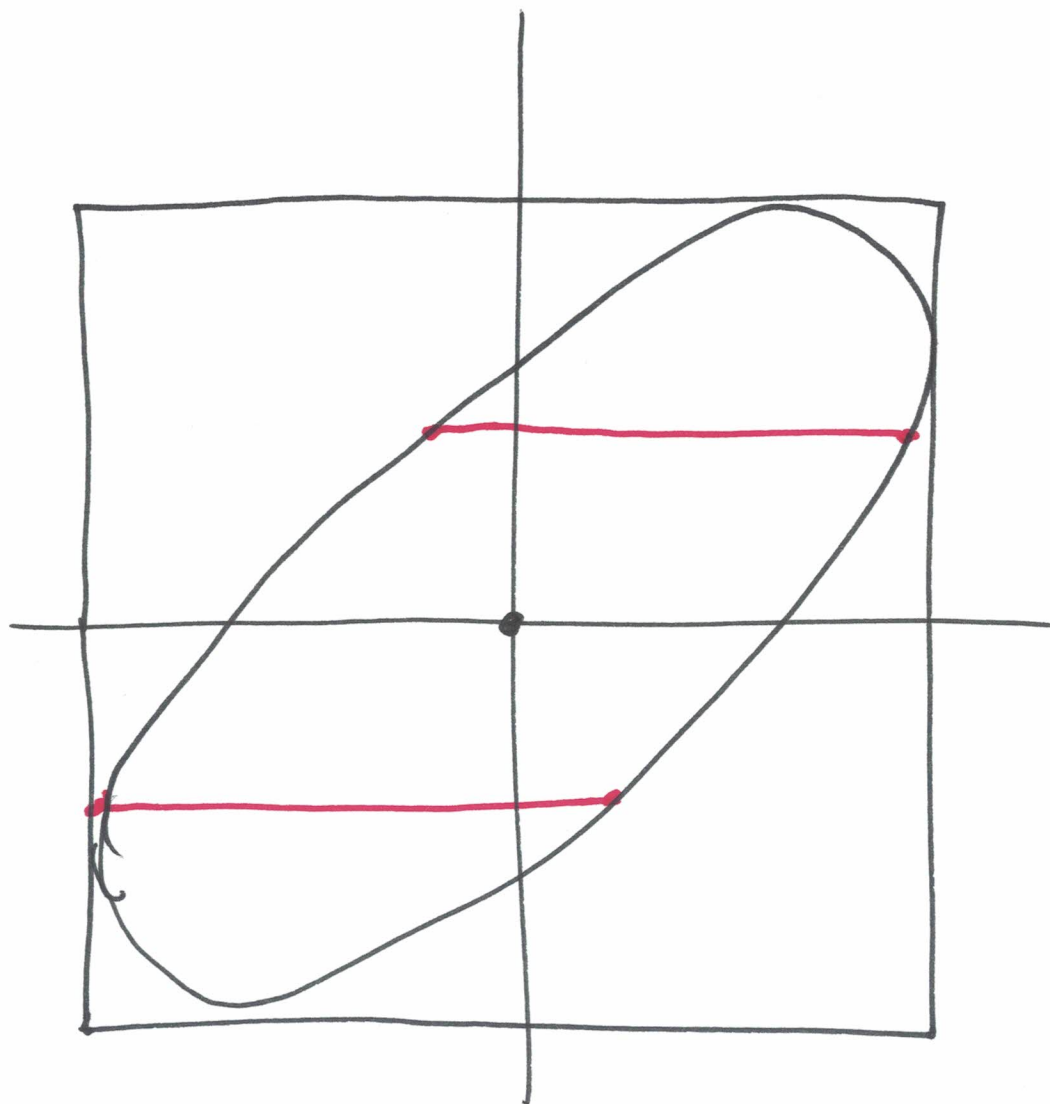
WHAT IF $\rho \neq 0$? ELLIPSE CONTOURS
ARE TILTED:

$$\rho = \frac{1}{2} \quad \sigma_x = \sigma_y = 1$$

~~$\sigma_x = \sigma_y = 1$~~ ~~$\sigma_x = \sigma_y = 1$~~

$$\mu_x = \mu_y = 0$$





ρ CAN RANGE BETWEEN -1 AND 1

SIDE
INFO

EX $\sigma_x = \sigma_y = 1$, $\rho = \frac{1}{2}$

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$$

SIZE, ORIENTATION OF ELLIPSE DEPENDS ON
EIGENVECTOR, EIGENVALUES OF THIS MATRIX.

$$A - \lambda I \left| \begin{array}{cc} 1-\lambda & \frac{1}{2} \\ \frac{1}{2} & 1-\lambda \end{array} \right| = \begin{array}{l} (1-\lambda)^2 - \frac{1}{4} = 0 \\ \lambda^2 - 2\lambda + \frac{3}{4} \end{array} \quad \frac{2 \pm \sqrt{4-3}}{2}$$
$$\left\{ \frac{3}{2}, \frac{1}{2} \right\} = 1 \pm \frac{1}{2}.$$

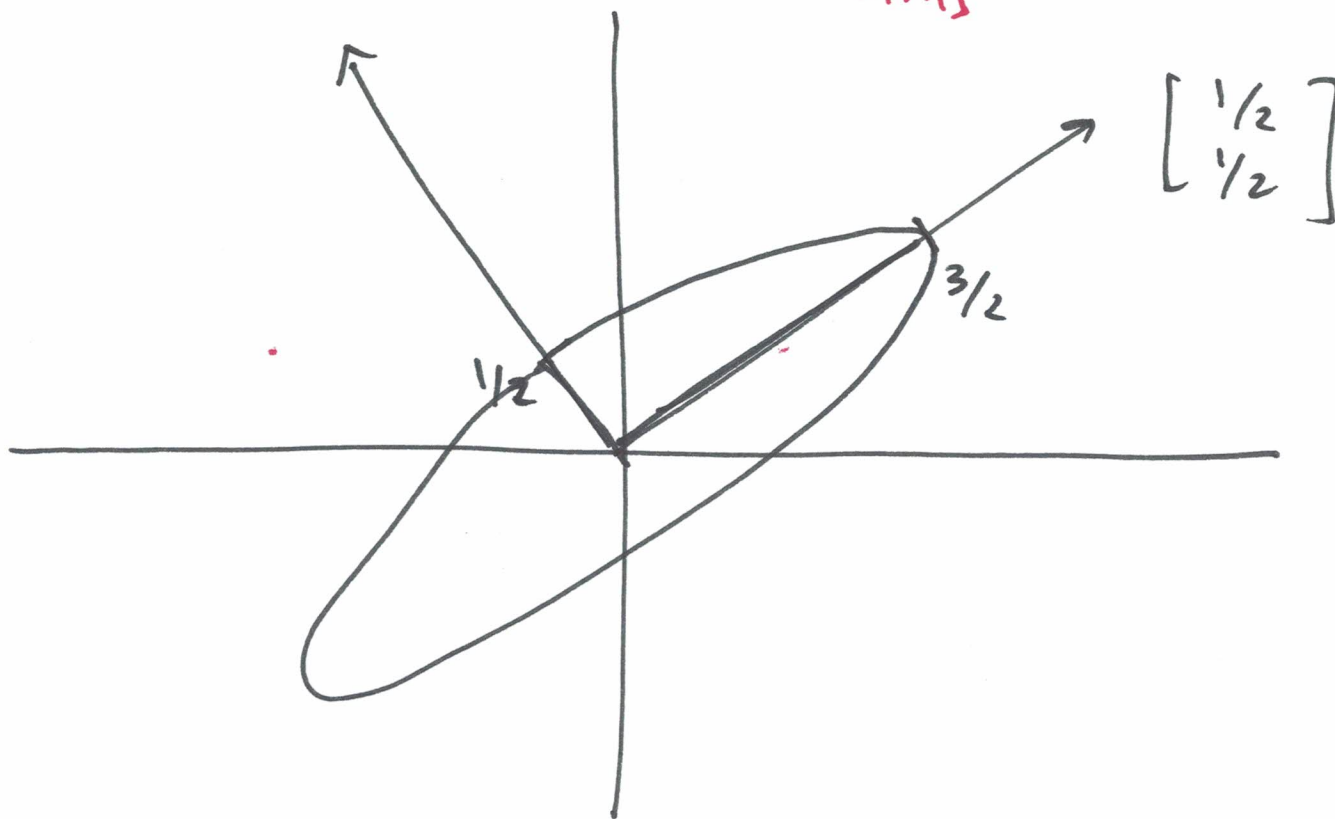
$$\lambda_1 = \frac{3}{2} \leftarrow \text{SIZE} \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$v_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\lambda_2 = \frac{1}{2} \Rightarrow v_2 \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

MAJOR AXIS
OF ELLIPSE

MINOR
AXIS



IN GENERAL $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ $\sigma_x = \sigma_y = 1$

$$\lambda = \{1 + \rho, 1 - \rho\}$$

$$\downarrow \quad \downarrow$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



$$\rho \rightarrow \underline{1}$$

RATIO $\frac{1 + \rho}{1 - \rho} \xrightarrow[\text{small}]{\text{Big}} \infty$

ρ CLOSE TO 1 $\frac{1.99}{0.01}$

GENERALLY

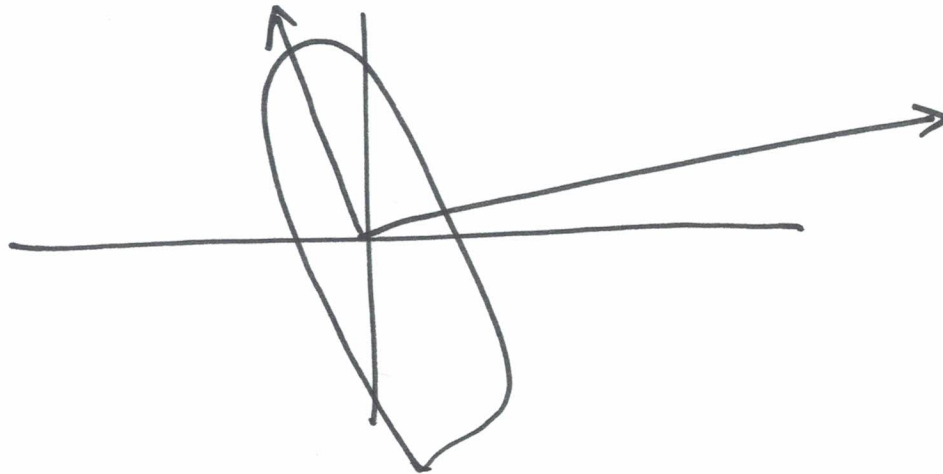
$$\sigma_X = 3$$

$$\sigma_Y = 5$$

$$\rho = -\frac{1}{3}$$

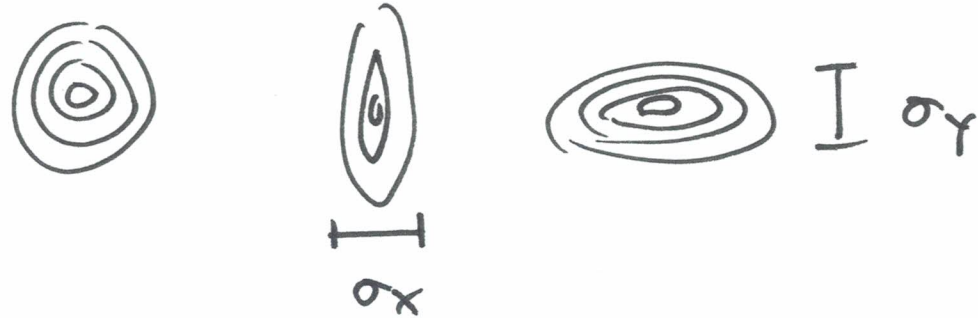
... $\lambda = [26.4, 7.5]$

$$\begin{array}{cc} \downarrow & \downarrow \\ \begin{bmatrix} -0.29 \\ 1 \end{bmatrix} & \begin{bmatrix} 3.49 \\ 1 \end{bmatrix} \end{array}$$

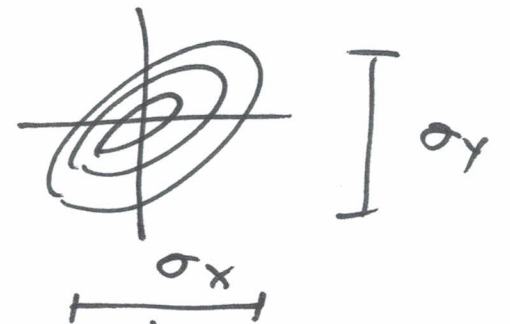


TAKE-HOME MESSAGE:

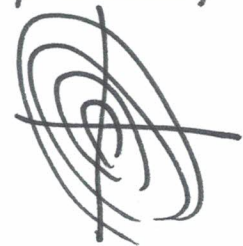
$\rho = 0 \Rightarrow$ CONTOURS ARE ELLIPSES ALIGNED WITH x, y AXES



$\rho > 0 \Rightarrow$ TILTED ELLIPSE LIKE



$\rho < 0 \Rightarrow$ TILTED ELLIPSE LIKE



$\rho \rightarrow \pm 1 \Rightarrow$ ELLIPSES ARE SKINNIER.

For HW 6:

$$E(g(x, y)) = \iint g(x, y) f_{xy}(x, y) dx dy$$